

**UCSC****University of Colombo, Sri Lanka***University of Colombo School of Computing***DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**Academic Year 2023 — 3rd Year Examination — Semester 6**IT6506 — e-Business Technologies***Structured Question Paper*
(2 Hours)**To be completed by the candidate****Index Number**

--	--	--	--	--	--	--

Important Instructions

- The duration of the paper is **2 hours**.
- The medium of instructions and questions is English. Students should answer in the medium of English language only.
- This paper has **4 questions** on **15 pages**. Answer **all** questions.
- All questions carry **equal** marks.
- Write your answers **only on the space provided** on this question paper.
- Do not tear off any part of this question paper. Under no circumstances may this paper (or any part of this paper), used or unused, be removed from the Examination Hall by a candidate.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **not allowed**.
- *All Rights Reserved*. This question paper can NOT be used without proper permission from the University of Colombo School of Computing.

**To be completed by
the examiners**

1	
2	
3	
4	
Total	

- 1) Read the following case study and answer the questions given below.

“BRICKS&TILES” is a leading bricks & tiles construction company that has been in the business for the last 30 years. BRICKS&TILES has contributed immensely to the construction industry in Sri Lanka, and it has also collaboratively worked with local and international construction companies, gaining expert knowledge and experience along the way.

BRICKS&TILES owner is Mr. Galagedara, who started by making clay bricks and cement tiles in his own backyard. He had the business skills and the ability to grow his business to make clay/cement bricks, many varieties of tiles, cement pots, concrete tiles and pots, etc. After 20 years in the business as a leader in providing bricks/tiles, BRICKS&TILES now has a good team of designers and constructors, including architects, surveyors, civil/mechanical/ electrical/ structural engineers, planning consultants, architectural consultants and many layers of workers. The company has also won many awards for its constructions as well as innovative solutions and designs.

BRICKS&TILES has a collaboration with a leading construction company, “BUILD”, which has been constructing single houses, housing schemes, commercial buildings, high-rise apartments, and bridges. BRICKS&TILES with BUILD was the largest and most experienced construction company in Sri Lanka, with an unmatched record.

The end of the war in Sri Lanka brought new hope to the northeast's construction sector and attracted a few new competitors with international reputations. These companies came with high technology and the latest machinery and equipment, which BRICKS&TILES is second to. The raw material suppliers found new customers as well. With the COVID-19 pandemic, political unrest and the recession in the last few years, the construction industry slowed down, and BRICKS&TILES lost its glory.

With the expansion of the road network and apartment concept in the cities in Sri Lanka, BRICKS&TILES, with the partnership of BUILD, is thinking of building new collaborations with related industries. There are many other industries that are doing well, such as tourism, hotel industry, manufacturing sector, and health sector, with the expansion and relocation of many government sector offices.

BRICKS&TILES has 8 branches in Kandy, Galle, Kurunegala, Negombo, Ratmalana, Matara, Jaffna, Anuradhapura and the head office in Colombo. The branches and the head office are linked with a basic integrated solution that manages the financial accounting side of the business.

During and after the COVID-19 pandemic and lockdowns, many industries and individuals got used to online ordering and payments. BRICKS&TILES needs a more comprehensive solution to manage this competitive environment. This company is looking at new technologies and collaborations to forge ahead in this industry. With the great image and the financial stability, it is not difficult to invest in eBusiness and new technological research and development.

- (a) Carry out a Situational Analysis for the company “BRICKS&TILES” to identify the Strengths, Weaknesses, Opportunities and Threats. Provide your answer in point form.

(10 marks)

ANSWER IN THIS BOX

Strengths	Weaknesses
• BRICKS&TILES” is a leading construction company & has been in business for 30 years. • The owner of BRICKS&TILES has a wealth of experience in the construction industry in Sri Lanka and the company. • He has worked collaboratively with International construction companies gaining expert knowledge and experience. • BRICKS&TILES has a good team of workers. • It has designers and constructors including architects, surveyors civil/ mechanical/ electrical/ structural/ fire protection engineers, planning consultants, architectural consultants and many layers of workers. • BRICKS&TILES has been constructing, single houses, housing schemes, commercial buildings and high-rise apartments. • It has 8 branches and head office in Colombo. • The company has won many awards for its constructions as well as innovative solutions. • BRICKS&TILES has a collaboration with a leading construction company, “BUILD”. • BUILD was the largest and most experienced construction company in Sri Lanka.	• These companies came with high technology and the latest machinery and equipment that BRICKS&TILES is second to- BRICKS&TILES has outdated or obsolete technology. • The branches and the head office are linked with a basic integrated solution which manages the financial accounting side of their business. • It needs a more comprehensive solution to manage, in this competitive environment. • The branches and the head office are linked with a basic integrated solution and no ebusiness solution. • BRICKS&TILES needs a more comprehensive solution to manage this competitive environment.
Opportunities	Threats
• The end of the war in Sri Lanka brought new hope to the construction sector. • BRICKS&TILES is thinking of building new collaborations with the related industries. • There are many other industries that are doing very well such as tourism, hotel industry, manufacturing sector, health sector • the expansion and re-location of many government sector offices. • In the North and the East, many hotel projects are going on giving new hope to the construction sector in Sri Lanka. • BRICKS&TILES is looking at new technologies and collaborations to forge ahead in the industry. • expansion of the road network and apartment concept in the cities in Sri Lanka brings hope. • company is looking at new technologies and collaborations to forge ahead in this industry.	• With the recession in the last few years, the construction industry slowed down and BRICKS&TILES lost its Glory. • End of War attracted few new competitors with international reputation. • These companies came with high technology and the latest machinery and equipment that BRICKS&TILES is second to. • The raw material suppliers found new customers as well. • During and after the COVID-19 pandemic and lockdowns, many industries and individuals got used to online ordering and payments.

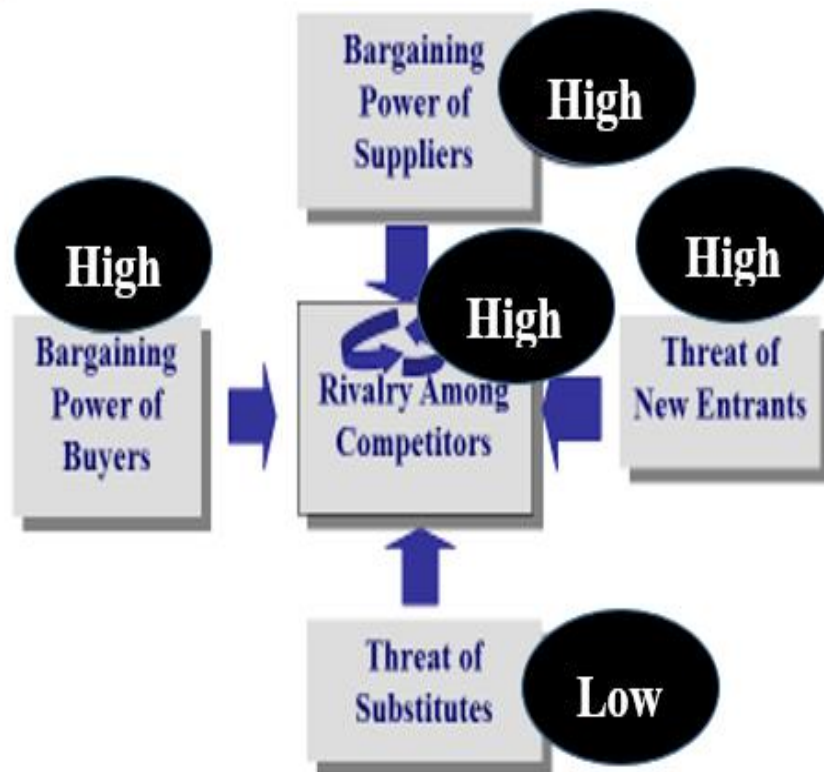
- (b) BRICKS&TILES needs to identify the environmental forces working on the company. For this purpose, illustrate and carry out a Michael Porter's Forces Analysis and Justify your answer.

(5 marks)

ANSWER IN THIS BOX

Michael Porter's Forces Analysis

Analyzing the External Environment

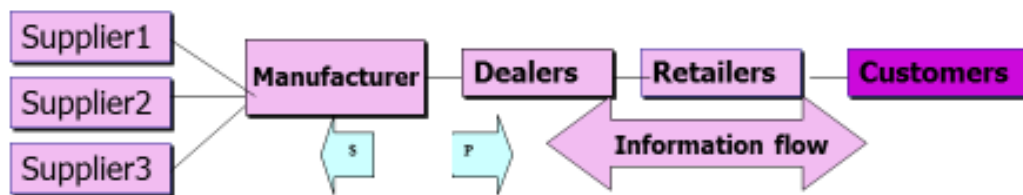


- (c) Propose two most suitable and beneficial eBusiness models for BRICKS&TILES to work collaboratively with BUILD and other companies. Illustrate and justify your selections. Also, write down eight (8) essential features to be included in the eBusiness solution.

(10 Marks)

ANSWER IN THIS BOX

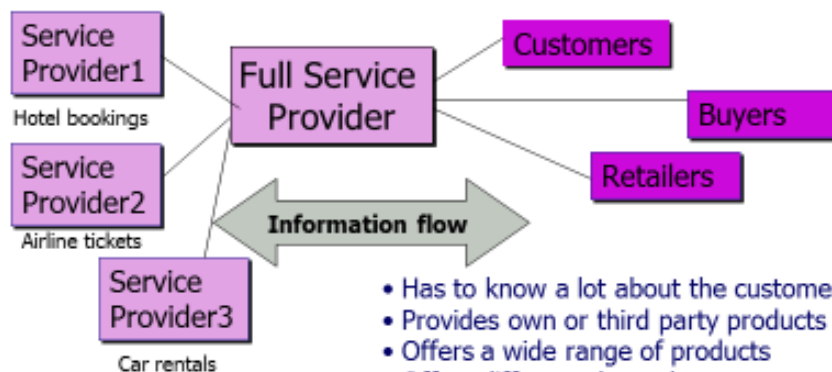
Supply Chain Model



- Virtual Value Chain
- Information flow across the supply chain
- All parties have a strong electronic bond and backend systems
- Some companies do/don't own any part of the value chain
- They have access to information about all from supplier/manufacturer to the customer



Full-Service Provider Model



- Has to know a lot about the customer
- Provides own or third party products
- Offers a wide range of products
- Offers different channels
Internet, face-to-face, phone, etc.
- Sells it's own products+ Commission for third party products
- Some charge customers a service fee

§

2) (a) Briefly describe the following with an example for each.

- (i) Price Disruption
- (ii) Product Disruption
- (iii) Mass Customisation
- (iv) Global Trade Platform for SMEs

(12 marks)

ANSWER IN THIS BOX

(i)

4.5. Price Disruption

"Price disruption" refers to a situation where a product or service is offered at a significantly lower price than the prevailing market price, often resulting in a price war or significant market share shift. Price disruption can occur for a variety of reasons, including new entrants with lower cost structures, increased competition, or changes in technology that make it possible to produce or deliver products more efficiently.

Price disruption can have a significant impact on businesses and industries. Companies that are able to adapt to lower prices and maintain profitability can gain a competitive advantage, while companies that are slow to adapt risk losing market share or becoming obsolete.

Examples of Price Disruption

Low-cost airlines: Low-cost airlines like Southwest, Ryanair, and EasyJet have disrupted the traditional airline industry by offering lower fares and no-frills service. By streamlining operations and minimizing costs, these airlines have been able to offer significantly lower prices than traditional carriers.

Generic drugs: Generic drugs have disrupted the market for branded pharmaceuticals by offering lower-priced alternatives. When a patent expires on a brand-name drug, other companies can produce and sell a generic version of the same drug at a lower price.

Online retailers: Online retailers like Amazon have disrupted traditional retail by offering lower prices and the convenience of online shopping. By operating with lower overhead costs and utilizing advanced logistics systems, online retailers can offer lower prices than brick-and-mortar stores.

Digital music: The rise of digital music disrupted the traditional music industry by allowing consumers to purchase and download individual songs or albums at a lower price than buying physical media. This has significantly reduced the cost of purchasing music and made it more accessible to consumers.

Solar energy: The cost of solar energy has been decreasing rapidly in recent years, disrupting the traditional energy industry by offering a lower-cost and renewable alternative to fossil fuels. As the cost of solar energy continues to decrease, it is likely to become an increasingly popular choice for consumers and businesses alike.

4.4. Product Disruption

Product disruption" refers to a situation where a new product or technology significantly impacts an existing product or market, often resulting in the obsolescence of older products or technologies. Product disruption can occur when a new product offers significant improvements in performance, convenience, or cost-effectiveness compared to existing products.

Product disruption can have a significant impact on businesses and industries. Companies that fail to adapt to product disruption can risk losing market share or becoming obsolete. However, companies that are able to innovate and develop new products that meet changing customer needs can often gain a competitive advantage and succeed in disrupted markets.

Product Disruption:

Examples of Digitization Across Industries

Industry	Traditional Format	Digital Format
Product Substitution		
Recorded music 	LP records, tapes	→ CDs, MP3
Journalism 	Newspaper, magazine, television, radio	→ Web site

2.4. Mass Customization

- Mass customization is a manufacturing process that combines the advantages of mass production and customization to create products that are tailored to the individual needs and preferences of customers, while still achieving economies of scale.
- In mass customization, products are designed and produced in a way that allows for a certain degree of flexibility and personalization, without sacrificing efficiency and cost-effectiveness. Customers are typically given a range of options to choose from, such as colors, sizes, features, and materials, and the products are then produced to their specifications using standardized components and processes.
- Mass customization is made possible by advances in technology, such as computer-aided design (CAD), 3D printing, and automation, which allow for the quick and efficient production of customized products. It has become increasingly popular in various industries, such as clothing, furniture, electronics, and automotive, as it enables businesses to differentiate themselves from their competitors and meet the unique needs and preferences of their customers.

2.4. Benefits of Mass Customization

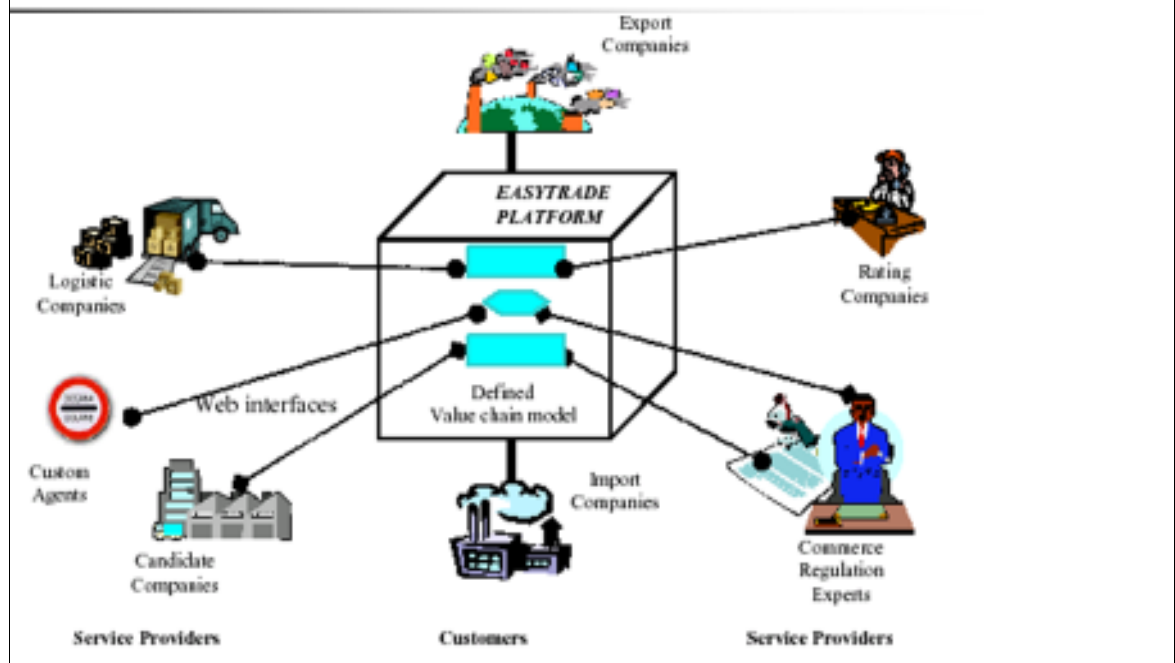
Mass customization offers several benefits for both businesses and customers, including:

- **Increased customer satisfaction:** Mass customization allows customers to personalize products according to their specific needs and preferences, leading to higher customer satisfaction and loyalty.
- **Higher revenue:** Customization often leads to higher product value, and customers may be willing to pay a premium price for products that are tailored to their needs, resulting in increased revenue for businesses.
- **Reduced inventory costs:** Mass customization can help reduce the need for businesses to hold large inventories of pre-made products, as products are only produced once they are ordered.
- **Faster time-to-market:** With standardized components and processes, mass customization can enable businesses to produce customized products faster than traditional made-to-order processes.

A Global Trade Platform for SMEs



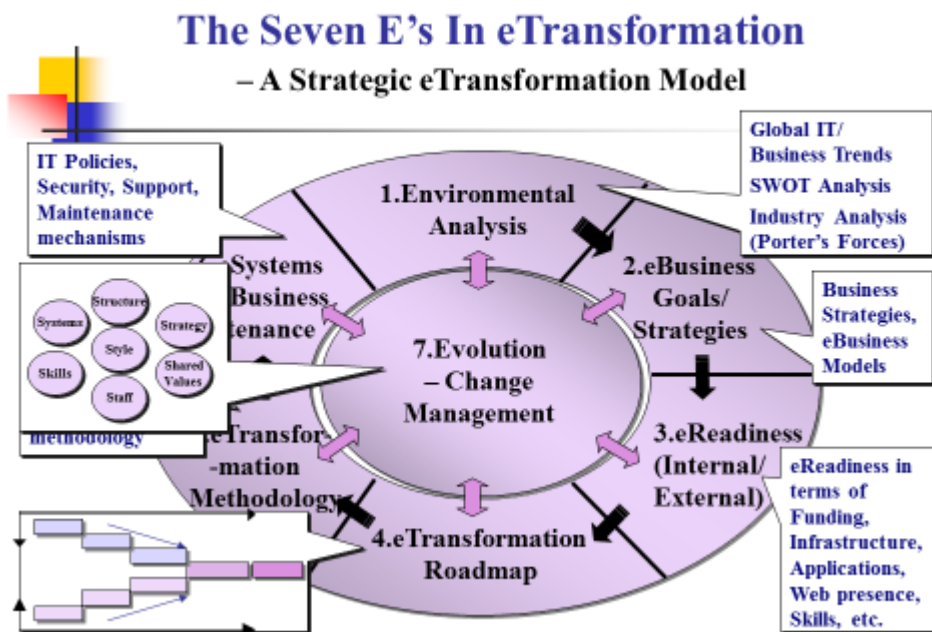
Global Trade Platform for SMEs



- (b) The Seven (7) Es of eTransformation is a formal methodology used to successfully take companies through the transformation process. Illustrate the 7Es in eTransformation and discuss the following stages.

- i. eReadiness
- ii. eTransformation Roadmap
- iii. eSystems

(7 marks)

ANSWER IN THIS BOX

i.

eReadiness (Internal/External)**Internal:**

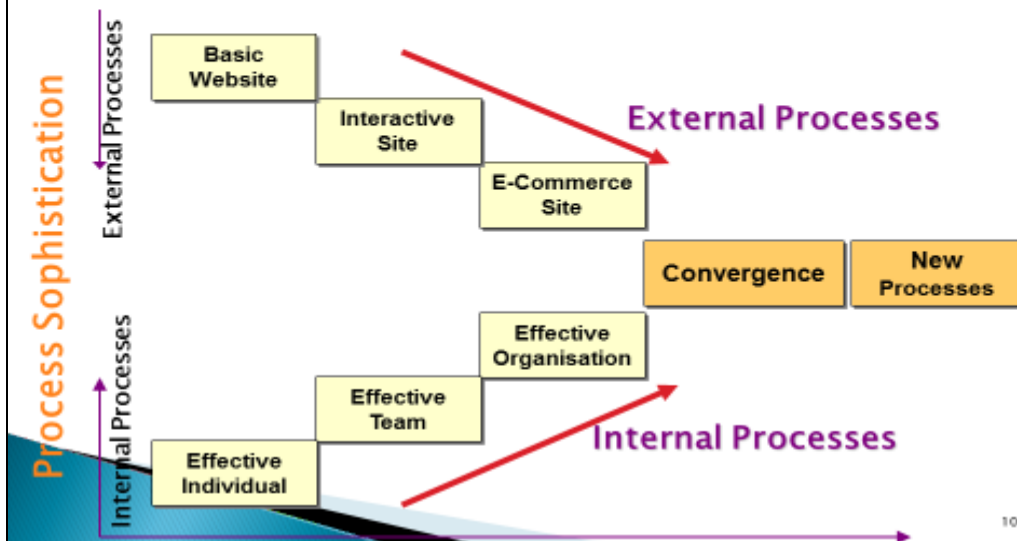
- Business processes – Well defined processes
- Applications & Infrastructure
- Web presence – Existence/ usage
- Skills –Level of IT skills of the employees
- Executive mgt – Commitment/Support
- External connectivity – Channels
- Future directions – Plans for expansions

External:

- Customers, Suppliers, Potential users

ii.

E-Business Road Map



10

iii.



eSystems (Policies, Support, Maintenance)



- **Management Controls :** Standards, guidelines to users, Procedures, Manuals
- **Security Measures :** To deal with common threats (sabotage, hacking, privacy, etc.) and contingency planning and disaster recovery
- **IT Maintenance and Support:** (Support for ICT infrastructure, upgrading, backing up, maintenance, troubleshooting, Support by the ISP and Vendors)

- (c) Describe an application where ICT could be effectively used in the Transportation Sector in Sri Lanka. Briefly discuss the following aspects.
- The problem being addressed
 - The eBusiness solution proposed
 - The challenges and benefits of the proposed solution

(6 marks)

ANSWER IN THIS BOX**ANSWER IN THIS BOX****Example: Designing a Traffic Management System (TMS) for Sri Lanka****i. Problem Being Addressed**

Traffic Congestion and Management Issues:

Sri Lanka faces significant challenges related to traffic congestion, especially in urban areas like Colombo and major highways. The inefficiencies in traffic management lead to increased travel times, fuel consumption, pollution, and frustration among commuters. There is also a lack of real-time traffic information available to drivers, to resolve these issues.

i. eBusiness Solution Proposed

Integrated Intelligent Traffic Management System: The proposed solution is an Integrated Intelligent Traffic Management System that leverages Information and Communication Technology to manage and optimize traffic flow effectively. Here are the components of the proposed system:

1. Real-time Traffic Data Collection:

- Implementing a network of sensors, cameras, and GPS trackers to collect real-time data on traffic flow, congestion levels, and road conditions.
- Using IoT (Internet of Things) technology to gather data from vehicles, traffic signals, and other infrastructure.

2. Centralized Traffic Control Center:

- Establishing a centralized Traffic Control Center equipped with advanced software and algorithms to analyze real-time data and make decisions.
- Integration of GIS (Geographical Information System) for mapping and visualization of traffic data.

3. Dynamic Traffic Management:

- Implementing adaptive traffic signal control systems that adjust signal timings based on current traffic conditions to optimize traffic flow.
- Providing dynamic rerouting suggestions to drivers through mobile apps, electronic signage, and GPS navigation systems.

4. Public Information Systems:

- Developing mobile applications and websites to provide real-time traffic updates, alternative routes, and travel advisories to commuters.
- Deploying electronic signage and displays at key junctions and highways to communicate traffic information effectively.

iii. Challenges and Benefits of the Proposed Solution

Challenges:

- **Infrastructure Development:** Building the necessary infrastructure such as sensors, cameras, and communication networks across the country.
- **Data Integration and Management:** Integrating data from various sources (e.g., vehicles, sensors) and ensuring its accuracy and reliability.
- **Technological Adoption:** Overcoming resistance to change and ensuring stakeholders (government agencies, private sector, and the public) adopt and use the new system effectively.

Benefits:

- **Reduced Congestion and Travel Times:** Optimizing traffic flow and reducing congestion through real-time monitoring and adaptive control systems.
- **Improved Air Quality:** Decreasing vehicle idle times and fuel consumption, which leads to reduced emissions and improved air quality.
- **Enhanced Safety:** Providing timely information to drivers about road conditions and hazards, thereby reducing accidents.
- **Less Frustration for Drivers:** Providing timely information to drivers about road conditions and hazards, thereby reducing crowded roads and idle time on the roads.
- **Efficient Resource Allocation:** Allocating resources (police, emergency services) more effectively based on real-time traffic data.
- **Improved Public Satisfaction:** Enhancing the overall commuting experience with reliable information and smoother traffic flow.

- 3) (a) Describe ride-hailing with an appropriate example.

(05 marks)

ANSWER IN THIS BOX

Ride-hailing is the process of hiring a taxi to travel from your current location to your destination, usually done through an app. In the past, people had to wait by the roadsides to catch a three-wheeler or call a cab service. Today, people enjoy the convenience of having their ride whenever needed. They call/hail a taxi/rider using a ride hailing app. There are two types of ride-hailing applications: Driver apps and rider apps.

Examples : Uber, PickMe

- (b) List down and briefly explain three (3) advantages of online banking.

(06 marks)

ANSWER IN THIS BOX

- Convenience
 - Customer can access their funds 24/7 from anywhere
 - There is no need to lose time waiting in line
- Fast & Efficient
 - Funds can be transferred within the same bank instantly
- Easy to monitor
 - Customers can easily monitor their transactions
 - Customers can get notifications as SMSes

- (c) Briefly explain how “Social Identity Theory” can be applied in digital marketing.

(05 marks)

ANSWER IN THIS BOX

o Create and promote brand communities - An online community can be created where customers with shared interests engage with each other.

It will increase the sense of belongingness.

o Use social proof and testimonials - Displaying ratings and reviews promote the social identity of the brand

o Highlight shared values and causes - Align your brand with social causes or values that resonate with your target audience.

- (d) In social media marketing, companies must consider various types of social sites. Describe three (3) types of social media platforms with appropriate examples, and explain how companies can use each of those platforms to interact with more customers.

(09 Marks)

ANSWER IN THIS BOX

Types of Social Sites a company need to consider

- Social Networks
 - e.g.- Facebook , Twitter
- Social Streaming
 - e.g. - YouTube, Vimeo
- Social Search
 - e.g. - Google, Yahoo
- Social Knowledge
 - e.g. - Wikipedia
- Social Blogging
 - e.g. - Company Blog, Company Forum
- Social Customer Service
 - e.g. - Company Customer Support, Twitter
- Social Publishing
 - e.g. - Niche Sites
- Social Bookmarking
 - e.g. - Stumbleupon, Pinterest

- 4) (a) "New technologies can be used to enhance Customer Relationship Management (CRM)." Justify this statement by providing two (2) examples of such technologies.

(08 Marks)

ANSWER IN THIS BOX

Artificial Intelligence/ Machine Learning

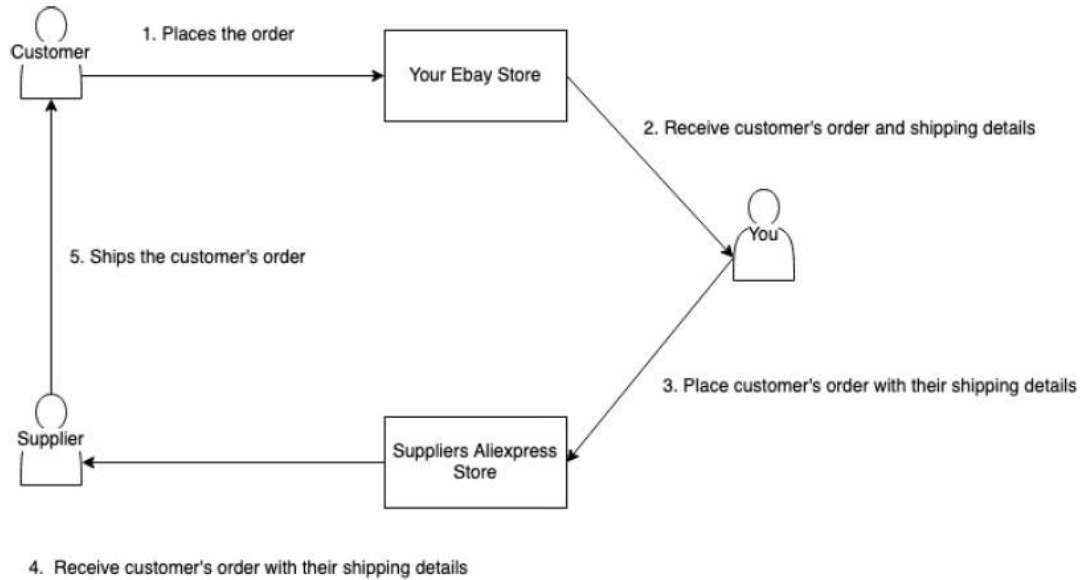
Chatbots

Virtual Reality and Augmented Reality

- (b) Illustrate the typical drop shipping scenario of an online marketplace using an appropriate diagram and label its steps.

(07 Marks)

ANSWER IN THIS BOX



- (c) “Fiverr is an ideal platform for freelancers who are just starting and have no prior client base.”
Is this statement true? Justify your answer.

(06 Marks)

ANSWER IN THIS BOX

Marks will be given for the explanation of advantages and the Justification.

Global Exposure

Diverse Range of Clients

Flexibility and Control

Portfolio Building

Additional Revenue Streams

Streamlined Workflow

Secure Payment System

Continuous Support

Ratings and Reviews

- (d) List down four (4) stages in the customer lifecycle.

(04 Marks)

<u>ANSWER IN THIS BOX</u>
Customer selection
Customer acquisition
Customer retention
Customer extension
